

Requirements Traceability Matrix

Human-Centred RE for Critical Systems — Validation Evidence for R01–R67

Legend: Supported Partially Supported Needs Revision Not Evaluated

Method: INT=Interview CW=Cognitive Walkthrough N/A=Not evaluated

R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
Elderly Communities (E1-E11)						
R01	E1: Reduce short-term memory load	Ensure the system simplifies navigation with a consistent and predictable structure.	NF (Usability)	Navigation described as easy to learn after a couple of attempts; bottom navigation bar felt familiar and predictable.	Partially Supported	INT/CW
R02	E1: Reduce short-term memory load	Ensure icons are clear, consistent, and semantically unambiguous across all screens.	NF (Usability)	Icons generally clear on home/guide screens; icons in Shelter in Place (wavy lines, fire extinguisher, shield) did not make sense to some users; icon clarity inconsistent across sections.	Needs Revision	INT/CW
R03	E1: Reduce short-term memory load	Ensure the system provides step-by-step instructions for key tasks.	NF (Usability)	Step-by-step guides praised; Take Action section described as “very clear... not too many choices”; elderly user highlighted simple language supported inclusivity.	Supported	INT/CW
R04	E2: Maintain focus on current action	Ensure the interface removes distractions and secondary functions, highlighting the most important or typical action on each screen.	NF (Usability)	Quick links to critical content on home page appreciated; most important actions were accessible up front.	Supported	INT/CW
R05	E3: Reduce options and available elements	Ensure the interface limits unnecessary choices during interactions to avoid overwhelming users.	NF (Usability)	Layout described as clean and simple; participants appreciated limited choices per screen.	Supported	INT/CW
R06	E4: Introduce features incrementally	Ensure the system introduces new features gradually to prevent overwhelming users with too many unfamiliar elements at once.	NF (Usability)	Onboarding described as gradual and confidence-building; elderly participant noted gradual familiarisation with the system.	Supported	INT
R07	E5: Readable text content	Ensure the system provides definitions or tooltips for abbreviations and jargon to improve text comprehension.	NF (Accessibility)	Plain language praised throughout; no tooltips for abbreviations present or tested in prototype.	Not Evaluated	N/A
R08	E6: Accuracy and precision	Ensure all interactive elements, including buttons and text, are consistently visible and legible across all screens.	NF (Accessibility)	Buttons described as appropriately sized; some text on home page described as too small; tension between compactness and legibility noted.	Partially Supported	INT/CW
R09	E7: Adaptable content presentation	Ensure the system supports alternative content presentations (e.g., audio or simplified layouts) without loss of information.	NF (Accessibility)	Audio and alternative layouts not implemented in Figma prototype; not evaluated.	Not Evaluated	N/A
R10	E8: Contextualised help	Ensure the system provides context-aware help automatically, reducing the need for users to search manually.	Functional	Help menu accessible via settings but not context-aware. No evidence that help appeared automatically during tasks.	Not Evaluated	N/A

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R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
R11	E9: Provide a safe exit on any screen	Ensure every process screen includes back, cancel, or exit buttons to let users exit at any point without losing control.	Functional	Quick links at end of pages praised; back navigation available but required extra effort in some cross-section flows.	Partially Supported	INT/CW
R12	E10: Don't assume knowledge of symbols	Ensure icons are intuitive and semantically meaningful; avoid abstract or ambiguous symbols.	NF (Usability)	Some icons described as leaving "no room for confusion"; others required thinking twice; Shelter in Place icons explicitly flagged as confusing.	Partially Supported	INT/CW
R13	E11: Prefer tapping over gestures	Ensure tapping alternatives exist for gesture-based interactions (e.g., sliders) to reduce dexterity demands.	Functional	Tap-based navigation used throughout; no gesture-based interactions present in prototype — requirement met by design.	Supported	INT/CW
Low-Digital-Literacy Communities (D1–D15)						
R14	D1: Include short, simple Help instructions	Ensure help menus provide concise, jargon-free, and easy-to-act-on instructions.	NF (Usability)	Adaptive menus described as easy to follow; some users skipped options because terminology was confusing (e.g. "Push" vs "Push Notifications").	Partially Supported	CW
R15	D2: Enable customisation	Ensure users can customise content and layout via settings to fit their needs.	Functional	Settings page found accessible; users appreciated control over colour scale, font size, and navigation mode; some colour customisation options not intuitive.	Partially Supported	INT/CW
R16	D2: Enable customisation	Ensure customisation options are self-explanatory without requiring external help.	NF (Usability)	Some users skipped options they "didn't understand"; "Push" terminology caused confusion; no confirmation feedback when changes were applied.	Needs Revision	INT/CW
R17	D3: Consistency	Ensure consistent colour schemes, typography, and navigation patterns across the entire application.	NF (Usability)	Layout consistency praised; app easy to learn due to predictable patterns; font "Inter" described as clean and consistent.	Supported	INT/CW
R18	D4: Predictable behaviour	Ensure the app behaves predictably, with navigation and interactions matching user expectations at all times.	Functional	Navigation generally predictable; inconsistent error text alongside a contact button created conflicting signals.	Partially Supported	INT/CW
R19	D5: Use multiple modes of interaction	Ensure content is available in multiple modalities (e.g., text, audio, video) for better comprehension.	Functional	Audio modality not implemented in Figma prototype; multi-modal content not evaluated.	Not Evaluated	N/A
R20	D6: Use distinct audio output	Ensure audio outputs are customisable and distinct so users can control volume and recognise meanings.	Functional	Audio not implemented in prototype; not evaluated.	Not Evaluated	N/A
R21	D7: Avoid unfamiliar visuals	Ensure visuals use universal or culturally appropriate symbols to improve recognition.	NF (Accessibility)	Icons generally understood; evacuation illustrations flagged as confusing ("what is the arrow?"); culturally specific symbols not explicitly tested.	Partially Supported	INT/CW
R22	D8: Avoid abstract and detailed imagery	Ensure visuals are semi-abstract and focused on key actions; avoid unnecessary detail.	NF (Usability)	Short text combined with visuals appreciated; some icons seen as superfluous; evacuation illustrations too abstract in places.	Partially Supported	CW
R23	D9: Maintain conceptual models	Ensure design uses affordances and signifiers to preserve a consistent mental model across the interface.	NF (Usability)	Bottom navigation bar aided mental model; cross-section navigation between danger-rating pages and guides broke mental model for some users.	Partially Supported	INT/CW

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R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
R24	D10: Provide affordances	Ensure interactive elements such as buttons and links are clearly distinguishable from static content.	NF (Accessibility)	Buttons described as distinguishable and appropriately sized; quick links easily tapped; no specific failures reported.	Supported	INT/CW
R25	D11: Apply constraints to reduce overload	Ensure the interface limits distractions by restricting interactions to only essential actions and respects cultural constraints.	NF (Usability)	Minimalist layout praised; limited choices per screen appreciated. Cultural constraints not explicitly tested.	Supported	INT/CW
R26	D12: Provide clear signifiers	Ensure app design communicates expected outcomes using visual/textual cues without relying solely on instructions.	NF (Usability)	Danger rating banners and guide tier colours provided clear signifiers; some users unsure what action was needed once a danger rating appeared.	Partially Supported	INT/CW
R27	D13: Natural mapping	Ensure controls reflect real-world relationships (e.g., volume sliders aligned with perceived loudness).	NF (Usability)	Volume controls not present in prototype; fire danger ratings mapped effectively to colour-coded guides.	Partially Supported	CW
R28	D14: Adopt culturally responsive design	Ensure visuals, colours, and symbols align with local cultural preferences.	NF (Accessibility)	CFA standard danger rating colours used; culturally specific preferences not directly tested beyond the Australian context.	Not Evaluated	N/A
R29	D15: Error-free operation	Ensure the app is stable and fails gracefully with clear recovery options instead of crashing.	NF (Reliability)	Figma prototype does not crash; graceful failure under real connectivity conditions not testable in prototype.	Not Evaluated	N/A
Rural Communities (R1–R5)						
R30	R1: Split content into pages	Ensure long content loads in smaller paginated chunks to improve usability under poor Internet conditions.	Functional	Page-based navigation option available in settings; rural participants appreciated page mode to avoid losing focus when scrolling long content.	Supported	INT/CW
R31	R2: Promote app/website visibility	Ensure content is properly indexed (e.g., by search engines) and organised into discrete, discoverable pages.	Functional	Content indexing is a deployment concern not evaluable via prototype. Discrete page organisation validated as useful.	Not Evaluated	N/A
R32	R3: Learnability	Ensure navigation is logical and minimal, requiring little prior knowledge or training.	NF (Usability)	Navigation described as easy to learn; rural participants found app convenient and practical; some needed a few attempts to understand cross-section navigation.	Partially Supported	INT/CW
R33	R4: Memorability	Ensure app structures are intuitive and memorable so users can easily recall information and actions.	NF (Usability)	Consistent layout aided memorability; participants representing rural personas sometimes lost track of location within danger-rating section navigation.	Partially Supported	CW
R34	R5: Rely on telecommunication	Ensure the app supports SMS or offline-friendly messaging for areas with poor Internet connectivity.	Functional	Offline functionality confirmed as a key strength by rural CW participants; important for emergency management contexts. SMS not directly tested.	Supported	CW
Colour-Blind Communities (C1–C10)						
R35	C1: Test against tools	Ensure the design is validated using WCAG and colour-blindness simulators.	NF (Accessibility/Compliance)	Colour-blind persona participants used a simulator during CW; WCAG tool validation not directly observed during evaluation but referenced in design process.	Supported	CW

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R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
R36	C2: Use more than colour to distinguish	Ensure redundant visual cues (e.g., icons, text, patterns) supplement colour-coded information.	NF (Accessibility)	Danger rating text labels alongside colour confirmed as useful; some participants found colour customisation unreliable without sufficient redundancy.	Partially Supported	CW
R37	C3: Use thicker lines or larger elements	Ensure lines, fonts, and elements are sufficiently thick/large for visibility.	NF (Accessibility)	Element sizes generally adequate; some text described as too small; line thickness not specifically evaluated.	Partially Supported	CW
R38	C4: Avoid reliance on colour-blind-friendly palettes	Ensure designs rely on contrast and redundant cues rather than searching for a universally friendly palette.	NF (Accessibility)	Customisable colour scale provided; participants noted reliable combinations were hard to find; reliance on user-selected palettes without guidance flagged as insufficient.	Needs Revision	CW
R39	C5: Underline links	Ensure all hyperlinks are underlined or clearly marked beyond colour alone.	NF (Accessibility)	Link styling not specifically evaluated in CW sessions.	Not Evaluated	N/A
R40	C6: Supplement colour coding with text	Ensure colour-coded statuses or alerts include text labels as backups.	NF (Accessibility)	Fire danger rating text labels used alongside colour; text backup confirmed effective in CW.	Supported	CW
R41	C7: Use patterns	Ensure patterns (e.g., stripes/dots) provide redundancy for colour-coded areas such as charts or maps.	NF (Accessibility)	Map danger zones shown via colour only; no patterns used; participants using simulator found zones ambiguous.	Needs Revision	CW
R42	C8: Alter designs to avoid conflicts	Ensure UI is proactively designed to avoid problematic colour pairings that reduce contrast.	NF (Accessibility)	Colour customisation feature exists; colour-blind participants found difficulty with reliable combinations; problematic default pairings not proactively avoided.	Needs Revision	CW
R43	C9: Keep it simple	Ensure the interface limits excessive colours and shades to avoid ambiguity.	NF (Usability)	Limited colour palette used; guide tier colours mapped to CFA standards; no excess shading reported.	Supported	INT/CW
R44	C10: Respect colour-blind preferences	Ensure designs respect typical user preferences (e.g., favouring blue for clarity) and avoid negative shifts in meaning.	NF (Accessibility)	Colour customisation allows preference-setting; participant suggested restricting users from assigning the same colour to multiple danger ratings — negative meaning shift risk identified.	Needs Revision	CW
Shared: Elderly + Colour-Blind (EC1–EC3)						
R45	EC1: Colour contrast between popups and background	Ensure popup backgrounds and foreground text meet high contrast ratios for readability.	NF (Accessibility)	Contrast in popups not specifically evaluated; general readability of foreground/background described as adequate.	Not Evaluated	N/A
R46	EC2: Design without colour to maintain luminance contrast	Ensure designs work in greyscale by testing contrast without chromatic elements.	NF (Accessibility)	Greyscale testing not explicitly conducted; colour simulator used but luminance-only test not reported.	Not Evaluated	N/A
R47	EC3: Maintain WCAG contrast ratios	Ensure the design complies with WCAG AA/AAA standards (e.g., 4.5:1 for text) and validates dynamically with tools like Google Lighthouse.	NF (Accessibility/Compliance)	WCAG compliance referenced in design intent; dynamic Lighthouse validation not demonstrated. Elderly users noted font contrast generally adequate.	Partially Supported	INT/CW
Shared: Elderly + Rural (ER1–ER8)						

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R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
R48	ER1: Give time to act and read	Ensure time-sensitive actions allow extended durations so slower readers are not disadvantaged.	Functional	No timed events or deadlines present in prototype; requirement met by design.	Supported	INT/CW
R49	ER2: Improve discoverability	Ensure screens provide clear focal points and consistent visual hierarchy.	NF (Usability)	Home and guide screens described as having clear focal points; visual hierarchy well supported; quick links prominent on every screen.	Supported	INT/CW
R50	ER2: Improve discoverability	Ensure the interface always makes the user's current position and available return paths clearly visible.	NF (Usability)	Users lost track of location navigating between danger-rating pages and guide sections; blue dot on map mistaken for town marker; no legend or breadcrumb in cross-section navigation.	Needs Revision	INT/CW
R51	ER3: Keep minimalist, simple interface	Ensure the interface maintains a minimalist layout, keeping only essential elements visible and avoiding unnecessary complexity.	NF (Usability)	Minimalist approach generally praised; in places compactness created legibility issues; font on home page described as too small by some CW participants.	Partially Supported	INT/CW
R52	ER4: Use linear navigation	Ensure navigation avoids tree-structured hierarchies, using simple forward-and-back flows.	NF (Usability)	Linear guide navigation praised; cross-section navigation between danger ratings and guide pages introduced unintended branching that confused users.	Partially Supported	INT/CW
R53	ER5: Make onboarding easy	Ensure onboarding processes are simple, brief, and welcoming to first-time users.	Functional	Initial setup described as straightforward; elderly participant noted gradual confidence-building; one user admitted skipping options they did not understand.	Partially Supported	INT/CW
R54	ER6: Make app purpose clear	Ensure new users can easily understand the app's purpose without prior context.	NF (Usability)	App purpose (bushfire early warning) clear to all participants from home screen; no confusion about overall purpose reported.	Supported	INT/CW
R55	ER7: Break down information	Ensure long or dense content is broken down across screens using short sentences and bullet points to reduce cognitive load.	Functional	Checklists and paginated guide content praised; short sentences appreciated; evacuation section noted as listing dangers without corresponding responses.	Partially Supported	INT/CW
R56	ER8: Provide text alternatives	Ensure all non-text content includes descriptive text alternatives (e.g., alt-text).	NF (Accessibility)	Alt-text not testable in Figma prototype; not evaluated.	Not Evaluated	N/A
Shared: Rural + Low-Digital-Literacy (RD1–RD5)						
R57	RD1: Minimise download times	Ensure multimedia elements are optimised or replaced with lightweight alternatives to improve load times on poor connections.	NF (Performance)	Performance under poor connectivity not testable in Figma prototype.	Not Evaluated	N/A
R58	RD2: Consider alternative information infrastructure	Ensure app design supports shared-device usage and alternative distribution channels where app stores are inaccessible.	Functional	Shared-device and alternative distribution not testable in prototype; not evaluated.	Not Evaluated	N/A
R59	RD3: Reduce phone resource usage	Ensure minimal CPU, memory, and battery consumption for low-end or older devices.	NF (Performance)	Device resource usage not testable in Figma prototype.	Not Evaluated	N/A
R60	RD4: Reduce data use	Ensure data-heavy components (e.g., videos, maps) are optimised for users with limited mobile data plans.	NF (Performance)	Data optimisation not testable in Figma prototype; map data source (Google Maps) referenced for future implementation.	Not Evaluated	N/A

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R#	Guideline	Requirement	Type	Validation Evidence	Status	Method
R61	RD5: Enable offline access	Ensure essential features remain accessible offline or with intermittent connectivity.	Functional	Offline functionality confirmed as a key strength by rural CW participants; important for emergency management contexts.	Supported	CW
Shared: Elderly + Rural + Low-Digital-Literacy (ERD1–ERD5)						
R62	ERD1: Simplify navigation structure	Ensure the app-level navigation hierarchy is flat, with no more than two levels of depth.	NF (Usability)	Bottom navigation kept to one level; guide sections added a second level; cross-section navigation created perceived depth beyond two levels for some users.	Partially Supported	INT/CW
R63	ERD1: Simplify navigation structure	Ensure users can always determine their location within the app and return to a previous screen without confusion.	NF (Usability)	Users navigating between danger-rating pages and guides frequently lost their position; rural and low-digital-literacy persona participants more affected; return path not always visible.	Needs Revision	INT/CW
R64	ERD2: Reduce alternative paths	Ensure single, consistent routes for completing tasks to minimise confusion.	NF (Usability)	Task routes generally consistent; some users found multiple entry points to guide content created ambiguity about the canonical path.	Partially Supported	INT/CW
R65	ERD3: Avoid Hamburger Menu	Ensure critical functions remain visible and avoid hiding navigation options behind expandable menus.	NF (Usability/Accessibility)	Bottom navigation bar used throughout; no hamburger menus present; critical functions immediately visible.	Supported	INT/CW
R66	ERD4: Provide informative & clear feedback	Ensure progress, error, and confirmation feedback is clear, accessible, and non-intrusive.	Functional	Absence of confirmation feedback when applying colour scale changes flagged; no feedback confirming checklist saves; participants requested “Your colour scale has been updated” type messages.	Needs Revision	INT/CW
R67	ERD5: Understandable error messages	Ensure error messages are concise, instructive, and reduce user anxiety during recovery.	Functional	Inconsistent error text alongside a contact button increased rather than reduced anxiety; error messaging not fully evaluated.	Needs Revision	INT/CW